

PHASE 1 – Brainstorming & Ideation

1.1 Problem Statement

Many people want to improve their fitness but do not know how to create a suitable workout plan. Professional fitness guidance may not always be easily available. Generic workout plans may also fail to consider individual goals, schedules, and preferences.

1.2 Proposed Solution

FitBuddy is proposed as an AI-based fitness plan generator. Users provide basic information and their fitness goals. The Gemini model processes the information and generates a personalized fitness plan.

1.3 Objectives

To develop an AI-powered fitness planning system.

To generate personalized fitness plans.

To use Gemini Generative AI for intelligent responses.

To provide easy-to-understand workout suggestions.

To save time in creating fitness plans.

To make basic fitness planning more accessible.

1.4 Target Users

Beginners

Students

Working people

Fitness enthusiasts

People looking for basic workout guidance

1.5 Key Features

User-friendly interface

User information collection

Goal-based planning

AI-generated workout suggestions

Personalized recommendations

Gemini model integration

Quick plan generation

PHASE 2 – Literature Survey / Research

2.1 Introduction

Fitness applications have become increasingly popular because they help users track and improve their physical activity. Traditional applications generally provide predefined workout plans.

Generative AI can provide a more flexible approach by generating responses according to user-provided requirements.

2.2 Existing System

Existing fitness applications commonly provide:

Predefined workout routines

Exercise libraries

Activity tracking

Calorie tracking

General fitness recommendations

2.3 Limitations of Existing System

Plans may not be highly personalized.

Users may have difficulty selecting suitable exercises.

Some applications require extensive manual input.

Generic plans may not match individual schedules.

Professional guidance may be expensive or unavailable.

2.4 Proposed System

FitBuddy uses Gemini Generative AI to generate fitness plans according to user requirements.

2.5 Technologies Researched

Artificial Intelligence

Generative AI

Google Gemini

Prompt Engineering

Web/Application Development

GitHub

2.6 Research Conclusion

The research indicates that Generative AI can be used to create flexible and personalized responses. FitBuddy applies this capability to fitness planning while keeping the interface simple for users.

PHASE 3 – Problem Definition & Solution Design

3.1 Detailed Problem Definition

Users often find it difficult to design a workout routine that matches their goals, available time, and fitness level. A system is therefore required to generate an understandable and personalized fitness plan from user inputs.

3.2 Proposed Architecture

User

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FitBuddy Interface

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User Information

Age

Fitness goal

Activity level

Available time

Preferences

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Prompt Generation

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Google Gemini Model

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AI Processing

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Personalized Fitness Plan

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User

3.3 Functional Requirements

The system should accept user information.

The system should accept fitness goals.

The system should send suitable prompts to Gemini.

Gemini should generate a fitness plan.

The generated plan should be displayed to the user.

The system should handle invalid or incomplete inputs.

3.4 Non-Functional Requirements

Easy to use

Fast response

Reliable

User-friendly

Maintainable

Scalable

3.5 Main Modules

Module 1 – User Input

Collects the required information from the user.

Module 2 – Prompt Generation

Converts user information into a suitable prompt.

Module 3 – Gemini AI

Processes the prompt and generates the fitness plan.

Module 4 – Result Display

Displays the generated plan in a readable format.

PHASE 4 – Development / Implementation

4.1 Technologies Used

Frontend: HTML, CSS, JavaScript (if these are what you actually used)

AI Model: Google Gemini

Programming Language: Python / JavaScript (use the one actually used in your project)

Development Environment: Google AI Studio / Gemini API

Version Control: GitHub

4.2 Implementation Process

Step 1

Create the FitBuddy user interface.

Step 2

Add fields for user information.

Step 3

Collect the user's fitness requirements.

Step 4

Create a prompt using the collected information.

Step 5

Send the prompt to the Gemini model.

Step 6

Receive the AI-generated response.

Step 7

Display the generated fitness plan.

4.3 Example User Input

Age: 20

Fitness Goal: General fitness

Activity Level: Beginner

Available Time: 30 minutes

Days per Week: 5

4.4 Example Prompt

Generate a beginner-friendly fitness plan based on the user's age, fitness goal, activity level, available workout time, and preferred number of workout days. Organize the response clearly and include suitable exercise suggestions, rest periods, and general safety guidance.

4.5 Expected Output

The system generates a structured fitness plan containing:

Warm-up

Workout activities

Suggested duration

Rest periods

Cool-down

General fitness guidance

Note: The AI output should be treated as general information, not a substitute for advice from a qualified healthcare or fitness professional.

PHASE 5 – Testing & Validation

5.1 Testing

Testing is performed to check whether the system accepts user inputs correctly and generates an appropriate response.

Test Case

Input

Expected Result

TC01

Valid user details

Fitness plan generated

TC02

Different fitness goal

Goal-specific plan generated

TC03

Beginner activity level

Beginner-friendly plan generated

TC04

Missing information

System requests/handles missing input

TC05

Invalid input

Appropriate validation

TC06

Multiple user requirements

Customized response generated

5.2 Testing Result

The system successfully processes user information and generates AI-based fitness suggestions. Different inputs produce different responses according to the requirements.

5.3 Validation

The generated plans are checked for:

Relevance

Readability

Completeness

Appropriate structure

Response generation

PHASE 6 – Documentation

6.1 Advantages

Easy to use

Personalized output

AI-powered

Saves time

Simple interface

Flexible recommendations

Can support different fitness goals

6.2 Limitations

AI-generated information may not always be perfect.

Internet access may be required.

The system cannot replace professional medical or fitness advice.

The quality of output depends on the information provided by the user.

6.3 Future Enhancements

Future versions could include:

Exercise video demonstrations

Progress tracking

User accounts

Workout reminders

Diet-planning support

Fitness history

Mobile application

Integration with wearable devices

More advanced personalization

6.4 Conclusion

FitBuddy demonstrates how Generative AI can be used to create personalized fitness plans. By using the Gemini model, the system can process user requirements and generate structured fitness suggestions. The project provides a simple example of applying Generative AI to a practical everyday problem.

PHASE 7 – GitHub / Deployment

README CONTENT

Writing

FitBuddy – AI Fitness Plan Generator Using Gemini Models

Project Overview

FitBuddy is an AI-powered fitness plan generator that uses Google Gemini models to generate personalized fitness plans based on user requirements.

Problem Statement

Many people find it difficult to create a fitness routine suitable for their individual goals, activity level, and available time. FitBuddy provides an AI-based solution for generating customized fitness suggestions.

Objectives

Generate personalized fitness plans.

Use Gemini Generative AI.

Provide an easy-to-use interface.

Generate fitness suggestions based on user requirements.

Demonstrate a practical application of Generative AI.

Features

User information input

Fitness goal selection

AI-powered plan generation

Personalized workout suggestions

Simple user interface

Gemini model integration

Technologies Used

Google Gemini

Generative AI

Prompt Engineering

HTML/CSS/JavaScript or the actual technology used

GitHub

Working

User enters personal fitness requirements.

FitBuddy collects the information.

The information is converted into a prompt.

The prompt is processed by the Gemini model.

Gemini generates a personalized fitness plan.

The result is displayed to the user.

Future Enhancements

Progress tracking

Workout reminders

Exercise videos

Mobile application

Wearable-device integration

Advanced personalization

Disclaimer

FitBuddy provides general fitness information generated using AI. It is not a substitute for professional medical or fitness advice.

PHASE 8 – Demonstration / Final Submission

PPT CONTENT

Slide 1 – Title

FitBuddy

AI Fitness Plan Generator Using Gemini Models

Name: [Your Name]

Course: BCA

College: JCT College of Arts and Science

Slide 2 – Introduction

AI is increasingly being used in health and fitness applications.

Users often need personalized fitness guidance.

FitBuddy uses Gemini Generative AI to generate fitness plans.

Slide 3 – Problem Statement

Difficulty creating personalized workout plans.

Generic plans may not suit everyone.

Professional guidance may not always be accessible.

Users need a simple planning solution.

Slide 4 – Objectives

Develop an AI-powered fitness planner.

Generate personalized plans.

Integrate Gemini.

Provide a simple interface.

Demonstrate the use of Generative AI.

Slide 5 – Proposed Solution

User Input → Prompt → Gemini AI → Fitness Plan

Slide 6 – Technologies

Google Gemini

Generative AI

Prompt Engineering

Programming language/framework actually used

GitHub

Slide 7 – System Architecture

User

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FitBuddy Interface

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Input Processing

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Gemini Model

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AI-Generated Fitness Plan

Slide 8 – Working

Enter user details.

Select fitness goal.

Submit information.

Generate prompt.

Gemini processes the prompt.

Display personalized plan.

Slide 9 – Testing

Valid input testing

Goal-based testing

Invalid input testing

Missing input testing

Output validation

Slide 10 – Advantages

Personalized

Easy to use

AI-powered

Time-saving

Flexible

Slide 11 – Future Scope

Mobile application

Progress tracking

Exercise videos

Wearable integration

Workout reminders

Advanced personalization

Slide 12 – Conclusion

FitBuddy demonstrates the practical use of Gemini Generative AI for fitness planning. The project provides personalized fitness suggestions based on user requirements through a simple AI-powered system.

Slide 13 – Demo

Live demonstration of FitBuddy

Show:

Home/interface

User input

Fitness goal

Generate button

Gemini response

Generated fitness plan

Slide 14 – Thank You

Thank You

Questions? :::



Demo video order

Record your screen in this order:

1. Open FitBuddy → 2. Enter details → 3. Select goal → 4. Generate plan → 5. Show Gemini output → 6. Explain the result → 7. Show GitHub repository.

For your final submission, you'll also need the actual project screenshots, GitHub repository, source code, and whatever SmartBridge specifically asks for in each phase. If you upload the SmartBridge project document here, I can match these contents to its exact 8-phase templates rather than making you guess where each item goes.